

GNSS Spoofing Attack Detection By IMU Measurements Through A Neural Network

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Abstract—Spoofing attacks against global navigation satellite system (GNSS) receivers are a serious threat to secure navigation, also in autonomous driving. Cars typically include, beyond the GNSS receiver, also an inertial measurement unit (IMU), whose data can be used to detect GNSS spoofing attacks. We consider a specific spoofing attack, with the spoofed trajectory that gradually diverges from the true trajectory, and we propose a spoofing detection method based on machine learning. First, a feature vector is designed, collecting the difference of two estimates of the device velocity, obtained from the GNSS receiver and the IMU. Then, a neural network (NN) is trained over a set of true and spoofed trajectories to detect the attack. We compare the proposed solution with an approximated Neyman-Pearson test and a literature reference direct comparison method, confirming the low error probabilities of our novel solution.

I. INTRODUCTION

Nowadays many services and activities rely on an accurate estimation of the position, typically provided by a global navigation satellite system (GNSS) receiver. Indeed, GNSS signals are used for navigation though on-board car navigation systems, but they are also used by unmanned aerial vehicles (UAVs), including quadcopters and drones. Autonomous driving also needs accurate navigation systems. However, due to the open nature of GNSS signals for civil use, it is easy to generate and transmit a fake GNSS signal to induce a desired fake position estimate at a victim receiver. This is a *spoofing* attack, which has been reported several times over the recent years [1], [2].

Several devices, including cars but also smartphones, beyond incorporating a GNSS receiver, have also an inertial measurement unit (IMU), which provides measurements of acceleration, orientation, and angular velocity. Although IMU data are not precise enough to replace GNSS navigation, it is much harder to hack the IMU than doing a GNSS spoofing attack. Then, the IMU measurements, still related to the device movement, can be used to detect GNSS spoofing attacks.

Two detection approaches are typically considered in the literature: either IMU and GNSS measurements are directly compared, or they are fused (to improve the position estimate) and spoofing is detected as an anomaly of the fusion process. Approaches in the first category, here denoted as *direct comparison methods* (DCMs), have been proposed in [3], where IMU

and GNSS acceleration estimates are compared, [4], which includes also the comparison of the rotation rate, and [5], where estimated positions are compared, also based on other IMU measurements. In the second approach, the IMU estimations are used to improve the precision of the GNSS estimate, and spoofing detection reveals anomalies in the fusion process. A relevant application of this approach is the innovation testing (IT) [6], where the fusion is performed by Kalman filter and variations of the statistics of the normalized innovation reveal possible attacks. In another solution, two Kalman filters may be used, designed assuming legitimate and spoofing conditions [7], whose output is then compared; lastly, in [8] spoofing is detected through sharp variations of the Kalman filter state variables. All described solutions either do not assume any statistics of the observed GNSS measurements under attack, or require complex computations (such as Kalman filtering) and fine tuning of threshold parameters.

In order to obtain a simpler solution, still based on IMU and GNSS measurements, and capturing the different statistics of these data under legitimate and attack conditions, in this paper we consider a machine learning approach. In particular, we focus on a specific and challenging attack, wherein the spoofer induces a trajectory that gradually diverges from the true path of the victim device, and we aim at detecting such attack by observing a window of past IMU and GNSS measurements. The detection is performed by a neural network (NN), suitably trained with measurements obtained from existing databases, and simulating the spoofing attack conditions.

Note that machine learning approaches have been already applied for spoofing detection. In particular, in [9] three features are extracted from the correlation output of the tracking loop signal of the GNSS receiver and used as input of the s K-Nearest neighbourhood algorithm. In [10], the correlator output is used to detect anomalies in the position reconstruction procedure due to multipath, that can be useful also under a spoofing attack. In [11], spoofing detection based on the observed signal-to-noise ratio (SNR) of the GNSS signal, and using machine learning approach has been considered. The victim devices are UAVs in [12], and features used for detection are the pseudo range, Doppler shift, and SNR. However, all these solutions are based only on the GNSS signal, while in our paper we consider as feature vector also the measurements obtained by the IMU.

The rest of this paper is organized as follows. In Section II

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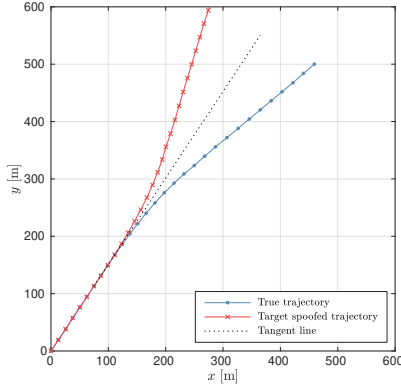


Figure 1. Example of true (blue line) and target spoofed (red line) trajectories used for numerical evaluation of the spoofing detection techniques. The target spoofed trajectory is the specular image of the true trajectory with respect to the tangent line (dotted line).

we introduce the system model, with a description of the considered attack, and a description of the IMU and GNSS signal models. Section III provides an overview of the spoofing detection by NN, with a description of the feature vector and the NN. A discussion on the training of the NN is also provided. Section IV describes two spoofing detection techniques then considered for comparison with our NN-based test: namely, the DCM and an approximated version of the Neyman-Pearson (NP) test. The detection error tradeoff (DET) curves obtained with our and the comparison methods are presented and discussed in Section V, before some conclusions are drawn in Section VI.¹

II. SYSTEM MODEL

We consider a *victim* device equipped with both a GNSS receiver and an IMU, including an accelerometer, a magnetometer, and a gyroscope. An attacking device transmits a suitable *spoofing* signal, decoded as GNSS signal by the victim device, to induce a false trajectory. Now, we first describe in more detail the attack and the characteristics of the IMU and the GNSS receiver.

A. Attack Model

We define as *trajectory* any continuous-time sequence of three-dimensional locations. In particular, we will focus on the the following trajectories:

- The *true trajectory*: the trajectory traveled by the device, also denoted as the ground truth trajectory;
- The *target spoofed trajectory*: the trajectory that the attacker aims at inducing at the victim GNSS receiver.

We will also denote as *legitimate* signal, the GNSS received when not under attack.

¹

Notation: Vectors and matrices are written in bold. The class of complex Gaussian random vector with mean $\mathbf{m}$ and covariance matrix $\mathbf{C}$ is denoted as $\mathcal{N}(\mathbf{m}, \mathbf{C})$. The class of chi-distributed random variables with n degrees of freedom is denoted as χ_n .

The attack that we aim at detecting is such that the target spoofed trajectory starts from a point of the true trajectory (where the spoofing attack starts), and then is the specular image of the true trajectory with respect to the tangent line. An example of attack is reported in Fig. 1. Note that this attack is challenging, because the spoofed position *slowly* diverges from the true position, while an attack introducing a sudden jump between the true and spoofed positions could be easily detected by thresholding the position measurements (or other related metrics).

In order to induce the target spoofed trajectory at the victim device, we assume that the attacker is capable of nulling the legitimate GNSS signals completely, such that, during the attack, the victim receives only the spoofing signal and locks onto it. Both the nulling and the spoofing signals can be easily generated by the attacker since the signal structure, together with the transmitted ephemeris data, are publicly available. Moreover, we assume that the attacker knows the position of the victim and is able to reproduce the legitimate signal at the victim (including delays of signals from various satellites due their relative the position of the victim), so that it nulls the legitimate signal at the victim by transmitting the legitimate signal with inverse polarity,

We also assume that the IMU is not affected by the attack, while still being subject to measurement errors as in nominal conditions.

B. IMU and GNSS Measurement Models

The IMU outputs measurements relative to the *body frame*, i.e., a coordinate system with origin and axes fixed to the device itself. GNSS measurements, instead, produces position estimates in earth-centered earth-fixed (ECEF) coordinates.

Our anti-spoofing technique will operate on the *navigation frame*, which has its origin at a reference point on the Earth's surface and its axes point towards North-East-Down (NED), respectively. Both IMU measurements and GNSS data are first translated into the navigation frame before anti-spoofing processing. The translation of GNSS measurements into NED coordinates is obtained via a fixed roto-translation of the ECEF coordinates. About IMU, we note that the relative orientation between the IMU frame and the navigation frame in general changes with time. However, efficient methods exist that estimate the device's attitude using IMU measurements, e.g., Kalman filters [13]. Once estimated, the device orientation can be used to rotate IMU measurements from the body frame to the navigation frame.

GNSS and IMU measurements are affected by additive time-correlated Gaussian noise [13], and, since rotations and translation do not change such error distribution, we assume that measurements are already expressed in the NED frame, which is also a common assumption in the literature [14].

The IMU measures acceleration at times nT_1 , providing the measurement

$$\tilde{\mathbf{a}}_n = \mathbf{a}(nT_1) + \mathbf{w}_{a,n}, \quad (1)$$

where $\mathbf{a}(nT_1)$ is the true device acceleration at time nT_1 in the navigation frame, and $\mathbf{w}_{a,n}$ is the Gaussian measurement noise. In particular, the noise has three independent zero-mean components, with covariance $\mathbf{r}_a$, i.e.,

$$\mathbf{w}_{a,n} \sim \mathcal{N}(\mathbf{0}_3, \mathbf{r}_a \mathbf{I}_3). \quad (2)$$

Similarly, GNSS measurements can be written as (see [15])

$$\tilde{\mathbf{p}}_k = \mathbf{p}(kT_G) + \mathbf{w}_{p,k}, \quad (3)$$

$$\mathbf{w}_{p,k} \sim \mathcal{N}(\mathbf{0}_3, \mathbf{r}_p \mathbf{I}_3), \quad (4)$$

where T_G is the GNSS sampling time, $\mathbf{p}(kT_G)$ is the true device position expressed in the navigation frame, and $\mathbf{w}_{p,k}$ is the additive Gaussian noise. Typically, the IMU sampling rate is higher than the GNSS receiver sampling rate, i.e., $T_G > T_1$.

III. SPOOFING DETECTION VIA NEURAL NETWORK

The purpose of spoofing detection is to decide between the two hypotheses

- $\mathcal{H}_0$: the GNSS receiver is not under spoofing attack, (5)
 $\mathcal{H}_1$: the GNSS receiver is under spoofing attack.

To this end, the basic idea is to compare IMU and GNSS measurements in order to detect inconsistencies in case of spoofing attack. This comparison is our case made by an NN. In particular, as typically done in the literature [3], [4], and [5], we aim at having a common measure obtained from the IMU and the GNSS receiver, to be compared: here we consider the velocity. Then, we compute the Euclidean distance between the two velocities, normalized by a suitable factor discussed in the following, and, unlike other works in the literature, we collect them into the *feature* vector, to be used as input to a suitably trained NN that provides as output a decision on whether GNSS is received is under attack or not. We now provide all the details of this procedure.

A. Velocity Computations

Starting from GNSS position measurements we perform numerical derivation, providing the velocity estimate at time kT_G as

$$\tilde{\mathbf{v}}_{G,k} = \frac{\tilde{\mathbf{p}}_{k+1} - \tilde{\mathbf{p}}_k}{T_G}. \quad (6)$$

We obtain the following additive error model $\tilde{\mathbf{v}}_{G,k} = \mathbf{v}_k + \mathbf{e}_{G,k}$, where $\mathbf{v}_k$ is the true velocity and $\mathbf{e}_{G,k}$ is the Gaussian noise vector which, from (3), can be written as

$$\mathbf{e}_{G,k} = \frac{\mathbf{w}_{p,k+1} - \mathbf{w}_{p,k}}{T_G}. \quad (7)$$

To compute the velocity from accelerometer measurements, we perform a trapezoidal approximation of the acceleration integral, i.e.,

$$\tilde{\mathbf{v}}_{I,n} = \tilde{\mathbf{v}}_{G,0} + \sum_{j=1}^n \frac{\tilde{\mathbf{a}}_j + \tilde{\mathbf{a}}_{j+1}}{2} T_1. \quad (8)$$

Again we obtain as error model $\tilde{\mathbf{v}}_{I,n} = \mathbf{v}_n + \mathbf{e}_{I,n}$, where, from (2), $\mathbf{e}_{I,n}$ is given by

$$\mathbf{e}_{I,n} = \mathbf{e}_{G,0} + \frac{T_1}{2} \sum_{j=1}^n (\mathbf{w}_{a,j} + \mathbf{w}_{a,j+1}). \quad (9)$$

B. Feature Vector

Defining $T_W = N_G T_G = N_I T_1$, the feature vectors collect the Euclidean distances between the GNSS and IMU velocities at sampling time T_G , within an observation window of duration T_W defined as

$$\mathcal{W} = [mT_G - N_G T_G, mT_G]. \quad (10)$$

Without loss of generality, in the rest of this section, we assume $m = N_G$, so that the window in (10) becomes $[0, T_W]$.

In order to build the feature vector, we obtain $\tilde{\mathbf{v}}_{I,k}^{(s)} = \tilde{\mathbf{v}}_{I,kZ}$ by down-sampling signal $\tilde{\mathbf{v}}_{I,n}$ at times kT_G by a down-sampling factor $Z = T_G/T_1$, which is assumed here to be an integer number.

Then, element k , $k = 0, \dots, N_G$, of the feature vector $\mathbf{f}(m) = [f_0(m), f_1(m), \dots, f_{N_G}(m)]$ is

$$f_k(m) = \frac{\|\tilde{\mathbf{v}}_{G,m-N_G+k} - \tilde{\mathbf{v}}_{I,m-N_G+k}^{(s)}\|}{\sqrt{\nu'_{m-N_G+k}}}, \quad (11)$$

where ν'_k is a normalization coefficient that depends only on $\mathbf{r}_p$, $\mathbf{r}_a$, T_G , and T_1 . The reason of normalization by ν'_k will be discussed in Section IV, while we here define it mathematically as $\nu'_k = \nu_p + \nu_k^{(s)}$, where

$$\nu_k^{(s)} = \nu_{kZ}, \quad (12)$$

$$\nu_n = \nu_p + \frac{(n+1)T_1^2}{2} \mathbf{r}_a(0), \quad \nu_p = \frac{2\mathbf{r}_p(0)}{T_G^2}. \quad (13)$$

C. Neural Network

The feature vector $\mathbf{f}(m)$ is then used as input to an NN. Recall that an NN can be described as a directed graph, where nodes are neurons and edges are links between them. Neurons are usually organized in one input layer, $Q - 2$ hidden layers, and one output layer. In particular, we used a feedforward network without cycles, i.e., one layer is connected only with the next one, and not with the previous one. Let $N^{(\ell)}$ be the number of neurons in layer ℓ . Thanks this structure, known as multilayer perceptron, an NN can be seen as a function mapping the feature vector into one real value.

Let $y_m^{(\ell)}$ be the output provided by neuron m of layer ℓ , and let collect all the outputs of layer ℓ into the vector $\mathbf{y}^{(\ell)} = [y_1^{(\ell)}, \dots, y_{N^{(\ell)}}^{(\ell)}]$. Neuron m of layer ℓ , $\ell = 0, \dots, Q - 1$, provides

$$y_n^{(\ell+1)} = \psi^{(\ell)}(\boldsymbol{\omega}_n^{(\ell)} \mathbf{y}^{(\ell)} + b_n^{(\ell)}), \quad (14)$$

where $\psi^{(\ell)}(\cdot)$ is the activation function, $\boldsymbol{\omega}_n^{(\ell)}$ is a vector of weights, and $b_n^{(\ell)}$ is a bias. A typical activation function is the logistic function, while both the weight vectors and the biases should be optimized in the learning phase in order to achieve the desired function with the NN.

In our setting, the input of the NN is the feature vector $\mathbf{f}(m)$, and the output is the scalar (thus $N^{(Q)} = 1$) $y_1^{(Q)}$. The spoofing test is performed by comparing $y_1^{(Q)}$ with a threshold λ , obtaining the test function

$$\hat{\mathcal{H}} = \begin{cases} \mathcal{H}_1 & \text{if } y_1^{(Q)} \geq \lambda, \\ \mathcal{H}_0 & \text{if } y_1^{(Q)} < \lambda. \end{cases} \quad (15)$$

The choice of λ will determine the DET between false alarm (FA) and misdetection (MD) probabilities,

$$P_{\text{FA}} = \mathbb{P}[\hat{\mathcal{H}} = \mathcal{H}_1 | \mathcal{H}_0], \quad P_{\text{MD}} = \mathbb{P}[\hat{\mathcal{H}} = \mathcal{H}_0 | \mathcal{H}_1]. \quad (16)$$

D. Training of the NN

Considering the specific attack we aim at detecting, as discussed in Section II-B, and the availability of error statistics in IMU and GNSS measurements, we can easily simulate features vectors both when receiving legitimate GNSS signals and when under attack. Indeed, in Section V, we will detail how such a training set has been built to obtain our numerical results. Therefore, we can resort to a supervised approach to learning the NN.

Let $\mathcal{S} = \{\mathbf{g}_i, i = 1, \dots, S\}$ be the set of S feature vectors and let $\mathcal{L} = \{l_i, i = 1, \dots, S\}$ be the set of corresponding labels, where $l_i = 0$ (1) means that the feature vector $\mathbf{g}_i$ has been obtained under legitimate (spoofed) conditions.

The empirical cross-entropy between the NN outputs and the true labels is

$$\mathcal{C} = -\frac{1}{S} \sum_{i=1}^S l_i \ln[y_1^{(Q)}(\mathbf{g}_i)] + (1 - l_i) \ln[1 - y_1^{(Q)}(\mathbf{g}_i)], \quad (17)$$

where $y_1^{(Q)}(\mathbf{g}_i)$ is the NN output in correspondence of the feature vector $\mathbf{g}_i$. The values of the biases and weights of the NN are adapted in order to minimize the empirical cross-entropy $\mathcal{C}$, using standard gradient descent algorithms.

Note that we used the cross-entropy as optimization metric, as it has been shown to provide asymptotic convergence of the hypothesis-testing NN to the optimal NP test, [16].

IV. COMPARISON TECHNIQUES

We introduce here two spoofing detection techniques that will be compared with our NN-based approach: one is an approximated version of the NP test, and while the latter is based on the direct comparison of measurements obtained by GNSS and IMU.

A. Approximated-Distribution Method

The most powerful test, minimizing the FA probability for a given MD probability, is the NP test, which however requires the knowledge of the probability density function (PDF) of the measurements under both hypotheses. Here we consider the NP applied on the feature vector $\mathbf{f}(m)$.

Under $\mathcal{H}_0$, since both $\tilde{\mathbf{v}}_{G,k}$ and $\tilde{\mathbf{v}}_{I,k}^{(s)}$ in (11) measure the same true velocity of the device, after the subtraction we are left with the norm of a Gaussian variable, which is chi-distributed. However, entries of $\mathbf{f}(m)$ are *correlated* since 1) noise measurement samples in (1) and (3) are time-correlated, and 2) derivation and integration in (7) and (9) generate additional correlation between the noise samples. Although correlation coefficients could be tediously computed, the resulting joint distribution of $\mathbf{f}(m)$ would be a correlated chi-distribution, for which closed-form expression do not exist, and even numerical evaluation is computationally unfeasible for higher dimensionality [17].

About $\mathcal{H}_1$, even considering the specific attack described in Section II, it is hard to obtain the PDF of the feature vector under attack, considering the various possible true and target spoofed trajectories.

Therefore, we consider an approximation of the NP test, where a) we ignore correlation under $\mathcal{H}_0$, and b) we assume not to know the statistics of the feature vector under $\mathcal{H}_1$. Thus, denoting with $\tilde{p}_{\mathbf{f}}(\mathbf{f})$ the approximated PDF of $\mathbf{f}(m)$ under $\mathcal{H}_0$, the approximated NP test is then

$$\hat{\mathcal{H}} = \begin{cases} \mathcal{H}_1 & \text{if } p_{\mathbf{f}}(\mathbf{f}) \leq \lambda_a, \\ \mathcal{H}_0 & \text{if } p_{\mathbf{f}}(\mathbf{f}) > \lambda_a. \end{cases} \quad (18)$$

The resulting test method is denoted as approximated-distribution method (ADM).

For the computation of $\tilde{p}_{\mathbf{f}}(\mathbf{f})$, we approximate the PDF of $\mathbf{f}(m)$ as the product of PDFs of chi distributions. Ignoring correlation, it can be easily shown that ν_n and ν_p in (13) are the noise powers of $e_{1,n}$ and $e_{G,k}$, respectively. Then, after down-sampling in (12), we obtain

$$\tilde{\mathbf{v}}_{G,k} - \tilde{\mathbf{v}}_{I,k}^{(s)} \sim \mathcal{N}(\mathbf{0}_3, \nu'_k \mathbf{I}_3). \quad (19)$$

Therefore, the normalization $\sqrt{\nu'_k}$ introduced in (11) is such that $f_k(m) \sim \chi_3$, and the approximate joint distribution of $\mathbf{f}(m)$ is

$$p_{\mathbf{f}}(\mathbf{f}) = \prod_{k=0}^{N_G-2} p_{\chi_3}(f_k), \quad (20)$$

with $p_{\chi_3}(\cdot)$ the chi distribution PDF with 3 degrees of freedom.

B. Direct Comparison Method

The DCM [4] is representative of a class of detection techniques directly comparing the positioning information coming from GNSS and IMU. DCM first interpolates the position measurements (3) from T_G to T_I , and then estimates the acceleration via double numerical derivation, i.e.,

$$\hat{\mathbf{a}}_n = \frac{\tilde{\mathbf{p}}_n - 2\tilde{\mathbf{p}}_{n-1} + \tilde{\mathbf{p}}_{n-2}}{T_I^2}. \quad (21)$$

The algorithm then compare $\hat{\mathbf{a}}_n$ with the accelerations measurements (1), by computing a component-wise correlation coefficient over the window $\mathcal{W}$ of (10). Following the notation of [4, eq. (8)], we have

$$\rho_i = \frac{\langle (\hat{\mathbf{a}}_n[i] - \langle \hat{\mathbf{a}}_n[i] \rangle_{\mathcal{W}})(\tilde{\mathbf{a}}_n[i] - \langle \tilde{\mathbf{a}}_n[i] \rangle_{\mathcal{W}}) \rangle_{\mathcal{W}}}{\sqrt{\langle (\hat{\mathbf{a}}_n[i] - \langle \hat{\mathbf{a}}_n[i] \rangle_{\mathcal{W}})^2 \rangle_{\mathcal{W}} \langle (\tilde{\mathbf{a}}_n[i] - \langle \tilde{\mathbf{a}}_n[i] \rangle_{\mathcal{W}})^2 \rangle_{\mathcal{W}}}}, \quad (22)$$

where $\langle x[i] \rangle_{\mathcal{W}}$ denotes the sample average over $\mathcal{W}$ and component i , $i \in \{1, 2\}$.² The detection metric and the test are now

$$\rho = \frac{1}{2}\rho_1 + \frac{1}{2}\rho_2, \quad (23)$$

$$\hat{\mathcal{H}} = \begin{cases} \mathcal{H}_0 & \text{if } \rho \geq \lambda_{\rho} \\ \mathcal{H}_1 & \text{if } \rho < \lambda_{\rho} \end{cases} \quad (24)$$

²We ignore the third component, since in our automotive experimental dataset (see Section V) we do not have meaningful vertical accelerations.

Table I
SENSOR PARAMETERS

Parameter	Value
GNSS position measurement variance σ_p^2	0.1 m ²
Accelerometer noise $\sigma_a \left(\frac{1}{\sqrt{2T_I}} \right)^{-1}$	$1.02 \cdot 10^{-5} \text{ m}/(\text{s}^2 \sqrt{\text{Hz}})$
Bias instability σ_{b_a}	$3.14 \cdot 10^{-3} \text{ m}/\text{s}^2$
Accelerometer decay factor ψ_a	0.5
GNSS decay factor ψ_p	0.8

where λ_ρ is a suitable threshold that guarantees a given false-alarm probability.

V. NUMERICAL RESULTS

We tested the performance of the proposed NN spoofing test over the public database [18] of recorded driving routes. In particular, we selected random segments of duration 45 s to be used as true trajectories, and we obtained the corresponding target spoofed trajectory as the specular image with respect to the tangent line, according to the attack described in Section II. The divergence instant is $t_A = 25$ s.

We evaluate the performance by considering the DET curves. The spoofing detection based on feature vector $\mathbf{f}(m)$ occurs at time $t_A + mT_G$, i.e., mT_G seconds after the two trajectories diverge.³

The NN has $Q = 3$ layers, and the activation function $\psi(x)$ is the identity at the input layer, the rectified linear unit (RELU) function at the hidden layer and the sigmoid function at the output layer. The training set size is 138 000, while the test is performed on 3 200 vectors.

Measurement Generation: For IMU and GNSS measurements generation we follow [13], thus $\mathbf{r}_a(\tau)$ and $\mathbf{r}_b(\tau)$ in (2) and (4) are given by

$$\mathbf{r}_a(\tau) = \frac{\sigma_{b_a}^2}{1 - \psi_a^2} \psi_a^{|\tau|} + \delta(\tau) \sigma_a^2, \quad (25)$$

$$\mathbf{r}_p(\tau) = \frac{\sigma_p^2}{1 + \psi_p} \psi_p^{|\tau|}, \quad (26)$$

where $\delta(t)$ is the Kronecker pulse and other parameters are specified in Table I. We also set the sampling times $T_G = 1$ s, $T_I = 10^{-2}$ s, and the sliding window duration $T_W = 25$ s: with this setting, the feature vector $\mathbf{f}(m)$ has 26 entries.

Performance Over Time: $d_{\min}$ is defined as the minimum divergence distance reached by a couple of true and spoofed trajectories within 10 s after t_A . Fig. 2 shows the DET of NN (solid lines), ADM (dashed lines), and DCM (dashed-dotted lines) for $d_{\min} = 3.75$ m and $m = 4, 8$, and 16. First, note that the gap between the NN and ADM is high. Indeed, the correlation of acceleration and position measurements makes the approximation of (20) not accurate. Next, we can see how the NN training takes advantage of the higher correlation and yields lower DET curves. Again, this justifies the use of

³This choice has only the purpose of showing detection performance under attack and both t_A and m are not needed as prior knowledge by the proposed detection algorithm.

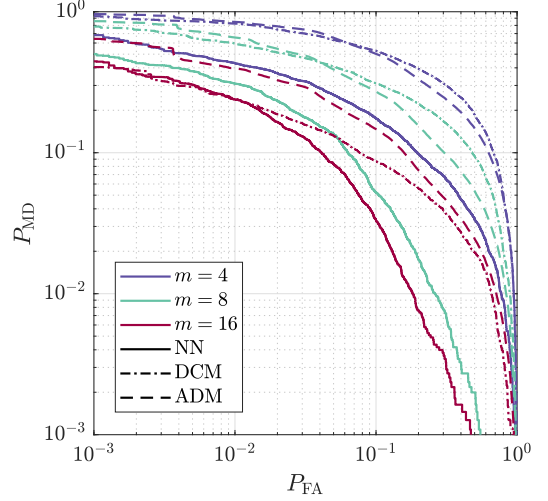


Figure 2. DET for the NN-based method (solid lines), ADM (dashed lines), and DCM (dashed-dotted lines), vs the detection instant m , with $d_{\min} = 3.75$ m.

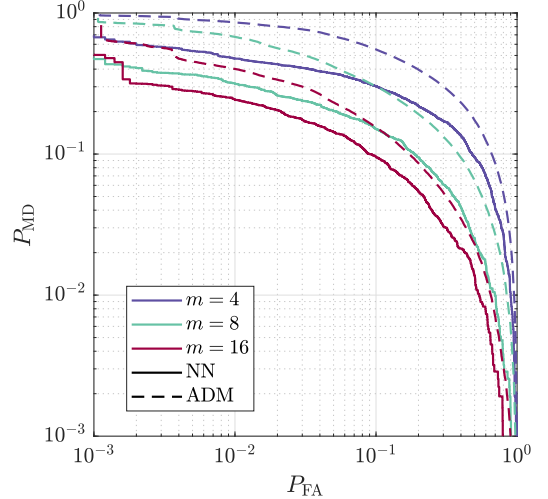


Figure 3. DET for the NN-based method (solid lines) and ADM (dashed lines) vs the attack detection instant m , while $d_{\min} = 3.75$ m for an uncorrelated measurement model.

machine learning techniques for spoofing detection, because NN training does not require any prior knowledge of the measurement distributions and adapts to diverse scenarios. Finally, the DCM yields worse performance in most cases, while performs like the NN for $m = 16$ and $P_{FA} < 10^{-2}$. However, note that the crucial instants are characterized by lower m because it is desirable to detect the attack as soon as possible. In these cases, i.e., $m = 8$ and $m = 4$, the proposed NN implementation always outperforms the DCM.

Contribution of Correlation of Measurements: For comparison purposes, we considered also a simplified measurements model without correlation of the estimation errors among the entries of $\mathbf{f}(m)$. In particular, we set the decay factors $\psi_a = 0$ and $\psi_p = 0$, such that the auto-covariance functions

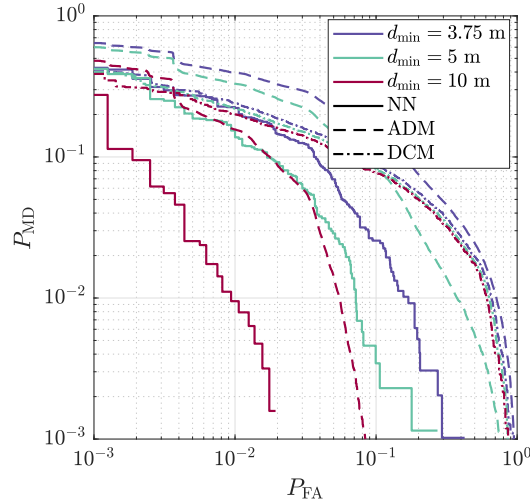


Figure 4. DET for the NN-based method (solid lines), ADM (dashed lines), and DCM (dashed-dotted lines), vs the minimum distance between trajectories $d_{\min}$, while $m = 16$.

(26) become $\mathbf{r}_a(\tau) = \sigma_a^2$, $\mathbf{r}_p(\tau) = \sigma_p^2$, yielding *uncorrelated* acceleration and position measurements. Note that the correlation introduced by (7) and (9) is still present. We then use the ADM and the NN-based spoofing detection technique via (15) and (18), and compute FA and MD probabilities (16). The NN is described in Section III-C, while we generated train and test sets comprising 144 000 and 10 000 trajectories, respectively. Fig. 3 shows the DET for the NN (solid lines) and the ADM test (dashed lines), vs the attack detection instant m , while $d_{\min} = 3.75$ m. We observe that the gap between the NN method and ADM is reduced, still the NN method outperforms ADM, as the first taken into account the statistics of the feature vector when under attack, and the correlation of the entries of the vector.

Impact of Trajectory Divergence: The same considerations can be made for Fig. 4, which shows the DETs of NN, ADM, and DCM for fixed $m = 16$ and different values of $d_{\min}$. Here the gap between the NN and the other two methods is even more evident, as the NN reaches $P_{\text{FA}} = P_{\text{MD}} \simeq 10^{-2}$ for $d_{\min} = 10$ m. This highlights once again the flexibility of the NN which, being asymptotically optimal, achieves better performance than any heuristic sub-optimal approach.

VI. CONCLUSIONS

In this paper we have presented a novel technique to detect spoofing attacks on GNSS signals, using the information coming from the IMU. The considered attack sees a gradual deviation of the target spoofed trajectory with respect to the true trajectory, thus being hard to detect. Numerical results has shown that our technique yields significantly lower FA and MD probabilities than DCM and another method based on the NP approach.

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